7.2	describe the structure of an atom in terms of protons, neutrons and electrons and use symbols such as ${}^{14}_{6}C$ to describe particular nuclei
7.3	know the terms atomic (proton) number, mass (nucleon) number and isotope
7.4	know that alpha (α) particles, beta (β^-) particles, and gamma (γ) rays are ionising radiations emitted from unstable nuclei in a random process
7.5	describe the nature of alpha (α) particles, beta (β^-) particles and gamma (γ) rays, and recall that they may be distinguished in terms of penetrating power and ability to ionise
7.6	*practical: investigate the penetration powers of different types of radiation using either radioactive sources or simulations*
7.7	describe the effects on the atomic and mass numbers of a nucleus of the emission of each of the four main types of radiation (alpha, beta, gamma and neutron radiation)
7.8	understand how to balance nuclear equations in terms of mass and charge
7.9	know that photographic film or a Geiger−Müller detector can detect ionising radiations
7.10	explain the sources of background (ionising) radiation from Earth and space
7.11	know that the activity of a radioactive source decreases over a period of time and is measured in becquerels
7.12	know the definition of the term 'half-life' and understand that it is different for different radioactive isotopes
7.13	use the concept of the half-life to carry out simple calculations on activity, including graphical methods
7.14	describe uses of radioactivity in industry and medicine
7.15	describe the difference between contamination and irradiation

Students should:	
7.16	describe the dangers of ionising radiations, including: • that radiation can cause mutations in living organisms • that radiation can damage cells and tissue • the problems arising from the disposal of radioactive waste and how the associated risks can be reduced

(c)	**Fission and fusion**
Students should:	
7.17	know that nuclear reactions, including fission, fusion and radioactive decay, can be a source of energy
7.18	understand how a nucleus of U-235 can be split (the process of fission) by collision with a neutron and that this process releases energy as kinetic energy of the fission products
7.19	know that the fission of U-235 produces two radioactive daughter nuclei and a small number of neutrons
7.20	describe how a chain reaction can be set up if the neutrons produced by one fission strike other U-235 nuclei
7.21	describe the role played by the control rods and moderator in the fission process
7.22	understand the role of shielding around a nuclear reactor
7.23	explain the difference between nuclear fusion and nuclear fission
7.24	describe nuclear fusion as the creation of larger nuclei resulting in a loss of mass from smaller nuclei, accompanied by a release of energy
7.25	know that fusion is the energy source for stars
7.26	explain why nuclear fusion does not happen at low temperatures and pressures, due to electrostatic repulsion of protons

8 Astrophysics

The following sub-topics are covered in this section.

(a) Units

(b) Motion in the universe

(c) Stellar evolution

(d) Cosmology